

About Me

My name is **Muhammad Ayan Rauf**. I am currently a *3rd semester* student of **BS Data Science** at FCIT, University of the Punjab, Quaid-e-Azam campus, Lahore. I belong to Sialkot, Punjab, Pakistan. Alongside my studies, I have a strong interest in football, cricket, chess, shooting volleyball, and solving maths problems. These interests connect well with the logical thinking needed in Data Science.

I am learning Data Science because I want to understand how data can be used to solve real-world problems. My future goal is to become an **AI/ML Engineer** as I am interested how models are trained to behave like humans and think on their own, or **Sports Tactical Analyst** because of my interest in sports. So far I have been working with **C++**, **Python**, **MySQL**, and **Excel**.

My Technology Skills

Technology	My Level	Experience	I use It For
C++	Intermediate	1 year	Data Structures Lab
MySQL	Intermediate	5 months	Querying, cleaning, and managing data
Excel	Intermediate	1 year	Data cleaning and Analysis
Python	Beginner	1 month	Data Analysis
Pandas	Beginner	Few weeks	Learning dataframe basics
NumPy	Beginner	Few weeks	Learning array operations
Power BI	Beginner	1 month	Building basic dashboards/reports
Markdown	Beginner	1 week	Writing formatted documentations and README files

My Learning Journey

1. Started learning *programming fundamentals* in **C++**
2. Studied *Object-Oriented Programming* in **C++**
3. Learning *Data Structures* in **C++**
4. Learned **MySQL** for querying and managing data
5. Learned **Excel** for data cleaning and analysis
6. Started learning **Python** for data analysis
7. Began exploring **Pandas** for working with dataframes
8. Began exploring **NumPy** for numerical operations
9. Started learning **Power BI** for building *dashboards*

My Learning Areas

- Programming
 - C++
 - Python
 - Data Science
 - MySQL
 - Excel
 - Pandas
 - NumPy
 - Data Visualization
 - Power BI
-

My Favorite Technologies

C++

I like C++ because it made me understand how programs actually work under the hood, especially through Data Structures and OOP. It helped me to understand memory allocation. I have used it in my Data Structures and OOP labs.

Resource: [C++ Reference](#)

MySQL

MySQL is one of my favorites because it allows me to query the data directly in a structured way. I have used it for practicing data querying exercises.

Resource: [MySQL Documentation](#)

Excel

I like Excel because it lets me clean, sort, and analyze data quickly through shortcut keys and formulas. I thought **Excel** was only for spreadsheets until I saw the dashboards I could build using it. I have used it for data cleaning and analysis tasks.

Resource: [Microsoft Excel Support](#)

Python

I am learning Python because it is the standard language for data science, and its simple, easy syntax lets me focus on solving problems instead of fighting with the code. I have not used it in a real project yet, but I've been practicing it.

Resource: [Python Official Website](#)

Power BI

I love Power BI because it turns raw data into visual dashboards, making insights way easier to explain. I'm currently using it to build basic reports and dashboards as a beginner.

Resource: [Microsoft Power BI Documentation](#)

Project Checklist

- ☒ Implemented **CString** class in **C++** using OOP
- ☒ Implemented **Stack** class in **C++**
- ☒ Implemented **Matrix** class in **C++**
- ☒ Implemented **Fractions** class in **C++**
- ☒ Implemented **ComplexNumbers** class in **C++**
- ☒ Cleaned and performed *EDA* on Nashville Housing dataset using **MySQL**
- ☒ Cleaned dataset and built dashboard for Superstore Sales dataset in **Excel**
- ☐ Completed a project in **Python**
- ☐ Completed a project in **Power BI**
- ☐ Completed an *end-to-end* Data Analytics project

Mathematical Expressions

Mean

$$\text{Mean} = \sum x / n$$

Standard Deviation

$$\text{Standard Deviation} = \sqrt{(\sum (x - \text{mean})^2 / n)}$$

Blockquotes

"In God we trust; all others must bring data." — W. Edwards Deming (*Statistician, Engineer, and Management Consultant*)

"Data is a precious thing and will last longer than the systems themselves." — Tim Berners-Lee (*Computer Scientist and Inventor of the World Wide Web*)

Image



These dashboards show how tools such as *Power BI* and *Excel* turn raw numbers into **visual insights** that are easy to understand at a glance.

Markdown Features I Practiced

- ☒ Headings
 - ☒ Bold
 - ☒ Italic
 - ☒ Inline Code
 - ☒ Ordered List
 - ☒ Unordered List
 - ☒ Nested List
 - ☒ Links
 - ☒ Images
 - ☒ Tables
 - ☒ Blockquotes
 - ☒ Mathematical Expressions
 - ☒ Checklist
 - ☒ Hard Line Breaks
 - ☒ Horizontal Rule
 - ☒ Strikethrough
-